



Python Programming - 2301CS404

Lab - 11

Roll No : 315

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Progrmas based on Python Data Structure + UDF + File Handling

Practical 1: Count Lines, Words and Characters

Problem Definition:

Write a Python program to read text from a file and count the total number of lines, words, and characters. The result should be written to an output file.

In [14]:

```
input_file = "abc.txt"
output_file = "output.txt"

line_count = 0
word_count = 0
char_count = 0

with open(input_file, "r") as file:
    for line in file:
        line_count += 1
        word_count += len(line.split())
        char_count += len(line)
```

```

with open(output_file, "w") as file:
    file.write(f"Number of lines: {line_count}\n")
    file.write(f"Number of words: {word_count}\n")
    file.write(f"Number of characters: {char_count}\n")

print("Counting completed. Results written to output file.")

```

Counting completed. Results written to output file.

Practical 2: Search a Word in a File

Problem Definition:

Write a Python program to search a given word in a text file and count its occurrences. Store the result in an output file.

```

In [15]: input_file = "abc.txt"
         output_file = "output.txt"

         word_occurrences = 0
         word = input('enter file words:')

         with open(input_file, "r") as file:
             content = file.read()
             words = content.split()

             for w in words:
                 if w==word:
                     word_occurrences += 1

         with open(output_file, "a") as file:
             file.write(f"Number of word_occurrences : {word_occurrences}\n")

         print("Counting completed. Results written to output file.")

```

Counting completed. Results written to output file.

Practical 3: Student Marks and Average

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Store marks in a list, calculate the average using a user-defined function, and write student details along with average marks to an output file.

```

In [16]: def calculate_average(marks):
         return sum(marks) / len(marks)

         with open("stu.txt", "r") as infile, open("output.txt", "a") as outfile:

             outfile.write("Roll No\tName\tMarks\t\tAverage\n")
             outfile.write("-----\n")

```

```

for line in infile:
    data = line.split()

    roll_no = data[0]
    name = data[1]
    marks = list(map(int, data[2:]))

    avg = calculate_average(marks)

    outfile.write(f"{roll_no}\t{name}\t{marks}\t{avg:.2f}\n")

print("Student details and average marks written to output.txt")

```

Student details and average marks written to output.txt

Practical 4: Maximum and Minimum using Tuple

Problem Definition:

Write a Python program to read numbers from a file, store them in a tuple, and find the maximum and minimum values. Write the result to an output file.

```

In [17]: with open("numbers.txt", "r") as infile:
        data = infile.read().split()
        numbers = tuple(map(int, data))

        maximum = max(numbers)
        minimum = min(numbers)

        with open("output.txt", "a") as outfile:
            outfile.write("Numbers: " + str(numbers) + "\n")
            outfile.write("Maximum value: " + str(maximum) + "\n")
            outfile.write("Minimum value: " + str(minimum) + "\n")

        print("Maximum and Minimum values written to output file.")

```

Maximum and Minimum values written to output file.

Practical 5: Remove Duplicate Words using Set

Problem Definition:

Write a Python program to read words from a file and remove duplicate words using a set. Store the unique words in an output file.

```

In [18]: with open("abc.txt", "r") as infile:
        data = infile.read().split()

        unique_words = set(data)

```

```
with open("output.txt", "a") as outfile:
    for word in unique_words:
        outfile.write(word + "\n")

print("Completed. Results written to output file.")
```

Completed. Results written to output file.

Practical 6: Word Frequency using Dictionary

Problem Definition:

Write a Python program to count the frequency of each word in a text file using a dictionary and write the result to an output file.

```
In [21]: input_file = open("input.txt", "r")
text = input_file.read()
input_file.close()

words = text.lower().split()

word_freq = {}

for word in words:
    if word in word_freq:
        word_freq[word] += 1
    else:
        word_freq[word] = 1

output_file = open("output.txt", "a")

for word, count in word_freq.items():
    output_file.write(word + " : " + str(count) + "\n")

output_file.close()

print("Word frequency count written to output.txt")
```

Word frequency count written to output.txt

Practical 7: Student Result and Grade Calculation

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Use a user-defined function to determine grade (A/B/C/Fail) and write the student result to an output file.

Grade Criteria:

- Marks ≥ 75 : Grade A
- Marks ≥ 60 : Grade B
- Marks ≥ 40 : Grade C

- Marks < 40 : Fail

In [27]:

```
def find_grade(marks):  
    if marks >= 75:  
        return "A"  
    elif marks >= 60:  
        return "B"  
    elif marks >= 40:  
        return "C"  
    else:  
        return "Fail"  
  
with open("stu.txt", "r") as infile:  
    lines = infile.readlines()  
  
with open("output.txt", "a") as outfile:  
    outfile.write("RollNo  Name  Marks  Grade\n")  
    outfile.write("-----\n")  
  
    for line in lines:  
        roll, name, marks = line.split()  
        marks = int(marks)  
  
        grade = find_grade(marks)  
  
        outfile.write(f"{roll:<7} {name:<6} {marks:<6} {grade}\n")  
  
print("Student results written to output file.")
```

Student results written to output file.

In []: